

Damian Lapena Vidal

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EDUCATION

University of Rhode Island (URI) | Kingston, RI Expected May 2027
Bachelor of Science in **Computer Engineering** GPA: 4.0/4.0

- **Scholarships:** URI Presidential, Subramaniam Family, C & W Eckman, Dr. & Mrs. E. Eddy, Barbara A. Woods Memorial
- **Courses:** Machine Learning I (Python/Scikit-Learn), Data Structures & Algorithms (C++), Electronics I (LTSpice), Linear Circuit Theory (MATLAB), Digital Circuit Design (NI Multisim), Digital Design with FPGAs (VHDL), Computer Architecture (MIPS)
- **Languages:** Spanish (Native), English (Fluent/Bilingual), German (Basic)

ENGINEERING EXPERIENCE

Student Technician | URI IT Services January 2026 – Present

- Using Q-SYS to monitor classroom tech across campus, making sure all hardware and digital interfaces are running smoothly
- Acting as the first point of contact for professors facing technical issues, providing quick on-site fixes or remote solutions
- Handling hands-on hardware work, from swapping out projectors to replacing cable infrastructure
- Work closely with a team of six to coordinate schedules and managing the ticketing system for all repairs

Quantum Computing Program | QuLearnLabs May 2025 – July 2025

- Learned principles of quantum mechanics relevant to computing, post-quantum cryptography, and quantum-safe protocols
- Designed and simulated quantum circuits using IBM Qiskit SDK
- Built a capstone project implementing Grover's Algorithm using 4 qubits

URI Undergraduate Research Assistant | URI Integrated Circuits Design Lab September 2024 – May 2025

- Assisted a team of three in PCB design using Altium Designer, incorporating transistors for cryogenic and radiation testing
- Performed manual soldering for prototype assemblies and ran standard condition simulations using LTSpice
- Conducted electrical characterization by measuring I-V curves using power supply and multimeter under standard conditions
- Developed a Python-based parameter optimization framework using PyTorch, integrating NGSpice for circuit simulation, finite-difference gradients for optimization, Git/GitHub for version control, and Weights & Biases for hyperparameter evaluation

PROJECTS

Low-Power Voice Assistant for Kids | URI College of Engineering June 2026 – Present

- Building a voice assistant that helps kids learn new languages through natural spoken conversation with friendly AI characters
- Engineering a real-time voice pipeline in async Python, streaming bidirectional audio between device and OpenAI Realtime API
- Delivered a working prototype on Raspberry Pi 5 and moving toward a lower power device designed for everyday use by kids
- Practicing AI product engineering from prompt architecture to voice model selection and safe AI behavior design

Campus AI Assistant | URI College of Engineering November 2025 – Present

- Developing a Retrieval-Augmented Generation (RAG) AI assistant to automate responses to high-volume student inquiries
- Using Python, LangChain, and OpenAI APIs to generate context-aware responses from institutional data
- Implementing real-time web scraping and preprocessing pipelines using Firecrawl and BeautifulSoup4 to collect university data
- Containerizing the system with Docker to support reproducible development and scalable deployment
- Collaborating in a 4-person team to design the retrieval pipeline, data ingestion workflows, and evaluation of AI responses

DigiSign | HackHarvard '25 (<https://digisign-6e9.pages.dev/>) October 2025

- Innovated a JavaScript-based e-signature capture tool that tracks a user's finger via webcam for secure payment transactions
- Built an interactive frontend using React, focusing on delivering smooth UX and an intuitive, user-friendly UI
- Developed backend services in Node.js, utilizing Google Gemini API for real-time signature processing and data storage
- Integrated Stripe API for safe wallet transactions, real-time payment verification, and automated billing management
- Handled deployment with Cloudflare and managed dependencies via Node Package Manager for efficient, scalable delivery
- Collaborated in a high-pressure team environment, leveraging strong communication, adaptability, and time management to deliver a functional MVP within the 36 hours hackathon deadline

LEADERSHIP EXPERIENCE

QuLearnLabs Ambassador February 2026 – Present

- Advancing QuLearnLab's mission to expand quantum literacy

URI Quantum Technologies Club Founder & President October 2025 – Present

- Founding the university's first student organization dedicated to Quantum Tech, focusing on the technology's future impact
- Leading a student executive board to handle the club's operations and help grow a meaningful quantum community
- Connecting with experts from industry and academia to invite guest speakers and give students a look at real-world careers

Hack@URI Organizer June 2025 – Present

- Helping in the planning, coordination, and behind-the-scenes work that brings a major tech event to URI
- Collaborating closely with other teams to manage expenses, organize reimbursements, and support sponsor follow-ups
- Attending leading hackathons such as HackHarvard to gather insights and implement innovative ideas for Hack@URI

CAMPUS INVOLVEMENT

Tau Beta Pi National Engineering Honor Society | Member December 2025 – Present

URI Investment Club | Member September 2025 – Present

URI Pickleball Club | Member September 2024 – May 2025

Society of Hispanic Professional Engineers | Member March 2024 – Present